Guided Inquiry Activity: 2D Steady-State Heat Conduction

Objectives

1. Understand the physical mechanism of steady-state heat conduction in two dimensions.
2. Explore the difference between **fixed temperature** (Dirichlet), **convective** (Robin), and **insulated** (Neumann) boundary conditions.
3. Investigate how material thermal conductivity (k) influences temperature profiles under different physical boundaries.
4. Analyze the physical significance of the dimensionless **Biot Number** (Bi) in heat transfer.

Background Physics

Steady-state heat conduction in a solid medium with constant thermal conductivity and no internal heat generation is governed by the **Laplace Equation**:

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

In our numerical simulator, we approximate this continuous partial differential equation using the **Finite Difference Method** on a structured grid. The interior temperature at any discrete node (i, j) is computed as the average of its four direct neighbors:

$$T_{i,j} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4}$$

However, the physical boundaries determine the unique solution to this equation. We model three types of energy balances at the boundaries:

- **Fixed Temperature (Dirichlet):** The edge is in perfect contact with a heat reservoir at a constant temperature $T_{boundary}$.
- **Insulated / Adiabatic (Neumann):** No heat escapes or enters the edge (heat flux $q'' = -k\frac{\partial T}{\partial n} = 0$).
- **Convective (Robin):** The edge is exposed to a moving ambient fluid at T_∞ with a heat transfer coefficient h. The surface energy balance is:

$$-k\frac{\partial T}{\partial n} = h(T - T_\infty)$$

Part 1: The Dirichlet Boundary Challenge (Fixed Temperatures)

When all boundaries of a system are held at fixed, constant temperatures, the steady-state profile is independent of the material itself. Let's test this phenomenon.

Procedures & Observations:

1. Open the simulator and select **Wood** as the preset material ($k = 0.15\ \mathrm{W/m \cdot K}$).
2. Configure all four boundaries to **Fixed Temp**:
   - **Top:** $100^\circ\mathrm{C}$
   - **Bottom:** $0^\circ\mathrm{C}$
   - **Left:** $50^\circ\mathrm{C}$
   - **Right:** $50^\circ\mathrm{C}$
3. Keep the **Mesh Size** at 45×45 and the **Initial Temp** at $25^\circ\mathrm{C}$.
4. Click **Start Simulation** and wait for the status to change to **Converged**. Note how many seconds it took to converge or observe the sweep speed.
5. Take a screenshot or sketch the steady-state temperature profile. Note the temperature in the geometric center of the slab.
6. Now, change the **Preset Material** to **Copper** ($k = 401\ \mathrm{W/m \cdot K}$). Do not change any boundary conditions.
7. Click **Start Simulation** and allow it to converge.

Inquiry Questions:

- **Q1.1:** Compare the final steady-state temperature profile of Wood and Copper under these fixed boundary conditions. Is there a difference in the color distribution or the temperature at the geometric center?
- **Q1.2:** From a mathematical standpoint, explain why the thermal conductivity (k) does or does not influence the final steady-state temperature distribution under pure Dirichlet (fixed temperature) boundary conditions. *Hint: Look at the Laplace Equation.*

Part 2: Introducing Thermal Resistance & Convective Boundaries

Now, let's explore what happens when we expose the slab to convective cooling, where the material's ability to conduct heat (k) competes with the fluid's ability to carry heat away (h).

Procedures & Observations:

1. Set the **Preset Material** to **Wood** ($k = 0.15\ \mathrm{W/m \cdot K}$).
2. Change the boundary conditions to:
    - **Top Boundary:** Fixed Temp, Value = $100^\circ\mathrm{C}$
    - **Bottom Boundary:** Fixed Temp, Value = $0^\circ\mathrm{C}$
    - **Left Boundary:** Convective, Ambient Value = $20^\circ\mathrm{C}$, $h = 45\ \mathrm{W/m^2 \cdot K}$
    - **Right Boundary:** Insulated
3. Click **Start Simulation** and wait for convergence. Note the temperature profile near the left (convective) and right (insulated) edges.
4. Next, change the **Preset Material** to **Aluminum** ($k = 237\ \mathrm{W/m \cdot K}$). Keep all boundary parameters identical.
5. Click **Start Simulation** and wait for convergence.

Inquiry Questions:

- **Q2.1:** Describe the shape of the temperature isotherms (lines of constant color) near the **Right Boundary** (Insulated) for both Wood and Aluminum. What physical law dictates that the isotherms must meet an insulated boundary at a 90° angle (perpendicular)?
- **Q2.2:** Look at the **Left Boundary** (Convective) for **Wood**. Does the edge of the wood slab actually reach the ambient fluid temperature of $20^\circ\mathrm{C}$? Why or why not?
- **Q2.3:** Contrast this with the convective edge of the **Aluminum** slab. Why does the high-conductivity material behave so differently at a convective boundary compared to the low-conductivity material?

Part 3: Mastering the Biot Number (Bi)

The **Biot Number** is a dimensionless quantity used in heat transfer calculations. It is defined as the ratio of conduction resistance inside the body to convection resistance at the surface of the body:

$$Bi = \frac{h \cdot \Delta x}{k}$$

Where Δx is the characteristic conduction length scale (here represented by grid cell spacing), h is the convection coefficient, and k is thermal conductivity.

Procedures & Observations:

1. Select **Custom Material** and set $k = 1.0\ \mathrm{W/m \cdot K}$.
2. Keep the grid mesh at 45×45.
3. Set the boundary conditions to:
    - **Top, Bottom, and Right Boundaries:** Insulated.
    - **Left Boundary:** Convective, Ambient Value = $0^\circ\mathrm{C}$, $h = 5\ \mathrm{W/m^2 \cdot K}$.
4. Set the **Initial Slab Temp** to $100^\circ\mathrm{C}$. This simulates a hot block of material exposed to cold air on its left side while being insulated on all other sides.
5. Click **Start Simulation** and pause or observe during the first few seconds of cooling.
6. Calculate the Biot number for this scenario (Bi_{low}).
7. Now, change the convective coefficient h on the Left Boundary to $500\ \mathrm{W/m^2 \cdot K}$.
8. Reset and run the simulation again. Observe the temperature gradient near the left edge.
9. Calculate the new Biot number (Bi_{high}).

Inquiry Questions:

- **Q3.1:** When $Bi \ll 1$ (low convection, high conduction), does the temperature inside the slab remain relatively uniform as it cools, or do steep temperature gradients form within the material? Explain the physical meaning of this regime.
- **Q3.2:** When $Bi \gg 1$ (high convection, low conduction), describe the temperature gradient near the convective boundary. How does the surface temperature compare to the ambient fluid temperature?
- **Q3.3:** If you were designing a heat sink, would you want a material/fluid interface characterized by a high Biot number or a low Biot number within the solid parts? Explain your engineering rationale.

Part 4: Numerical Methods and Initial Guesses

Let's analyze the behavior of the computational solver (Jacobi iterative solver).

Procedures & Observations:

1. Set the boundaries back to:
    - **Top:** Fixed, $100^\circ\mathrm{C}$
    - **Bottom:** Fixed, $0^\circ\mathrm{C}$

    - **Left:** Fixed, 50°C
    - **Right:** Fixed, 50°C
2. Choose **Wood** as the material.
3. Test two different **Slab Init Temperatures** (Initial Guesses):
    - **Test A:** 0°C
    - **Test B:** 100°C
4. Run each test and note how the heat "spreads" from the boundaries through the iterations.

Inquiry Questions:

- **Q4.1:** Did the initial guess alter the final, steady-state converged solution? Explain why or why not based on mathematical uniqueness of elliptic partial differential equations.
- **Q4.2:** Why is choosing an initial guess close to the expected average boundary temperature computationally advantageous?